A Smart Insole for Remote Gait Monitoring of Paralyzed Patients

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***Abstract*—Paralysis occurs when the nervous system is disrupted, preventing patients from making voluntary movements. The most common causes of paralysis are stroke and spinal cord injuries. Paralysis can result in a partial or complete loss of control over movement, often leading to an unsteady gait. Side falls, a severe consequence of this imbalance, frequently cause broken bones and head injuries. Real-time gait monitoring with integrated analysis could greatly improve the rehabilitation process for such patients. The proposed smart insole is designed to provide real-time and remote gait monitoring for paralyzed individuals. This innovative approach holds promise for enhancing personalized rehabilitation programs and enabling early detection of impaired movement and balance control.**

This study will assess various locomotion scenarios to detect potential gait impairments and early signs of balance loss. A smart insole, featuring multiple sensors precisely designed and strategically placed, is developed to measure key gait parameters. A proof-of-concept model of the insole has been created to conduct experimental tests and demonstrate its functionality in different human activities. The model includes a 3D-printed insole equipped with a nano Arduino controller and integrated gyroscopes and accelerometers embedded in the insole. Additional sensors are positioned at specific locations to measure other critical parameters. A Fiber Bragg Grating (FBG) sensor array is embedded in the insole to provide accurate, real-time detection of foot deformation, allowing for the identification of imbalances.

Keywords— Smart insole, Sensor Technology, Remote Gait Monitoring, Arduino, FBG.

I. Introduction

Stroke is one of the most common health issue worldwide, particularly among elderly people. Every year, over 16 million people worldwide suffer from strokes [1]. People experiencing stroke experience some level of paralysis, either temporary or permanently. Paralysis can result in a partial or complete loss of control over movement, often leading to an abnormal gait pattern. Loosing balance and foot drops are common consequences of loosing movement control. Patient with footdrop exhibit a pattern of motion compensation that causes the slowing down of swing rhythm. Deteriorating balance is one of the earliest signs of foot sensory loss in diabetic peripheral neuropathy (DPN), associated with a high rate of trauma due to falls and fear of falling [1]. Falls are a significant concern for adults experiencing sensory perception loss, with most incidents occurring during weight-bearing activities and being preventable. In addition to stroke, other disease such as diabetes may cause some sort of sensory perception loss [2].

A variety of ankle-foot orthoses have been designed to mitigate the effects of impaired locomotion and enhance the patient gait [3]. With advancement of sensor technology and wireless communications, in the past few years, attempts have been performed to develop smart wearable devices for precise gait monitoring.

Various smart insoles have been developed to provide a detailed analysis of a patient's locomotion. These high-tech inserts, designed to fit inside any shoe, monitor and collect data on movement and pressure distribution. One of their key advantages is the ability to incorporate multiple sensors without compromising the user's mobility [5]. Equipped with miniaturized sensors, smart insoles offer a promising solution for unobtrusive monitoring of daily activities[6-9]. By measuring plantar pressure, they provide valuable insights into foot and ankle function during gait and other activities, as these areas play a crucial role in weight-bearing and weight-shifting [10]. However, to the best of the authors' knowledge, many existing smart insoles suffer from limitations, including inaccurate measurements, a restricted range of monitored parameters, and over-specialization for specific conditions.

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This work aims to design an affordable, user-friendly smart wearable foot device to improve gait assessment for diagnosing, real-time monitoring, and prognostication in paralyzed patients. The insole is composed of a miniature control board, multiple sensors to detect acceleration and rotation angles, and Fiber Bragg Grating (FBG) sensors to measure foot deformation under various locomotion patterns. A proof-of-concept prototype of the smart insole has been developed, and several walking simulations have been conducted to demonstrate the system's performance and functionality across different walking scenarios.

II. DESIGN SMART INSOLE

A. Prototype smart insole

The 3D model of the insole has been designed with SolidWorks 2024-R1 and printed with a 3D printer (MakerBot). A miniature control with built-in accelerometers, gyroscope, and Bluetooth was embedded in the insole as shown in Fig. 1.

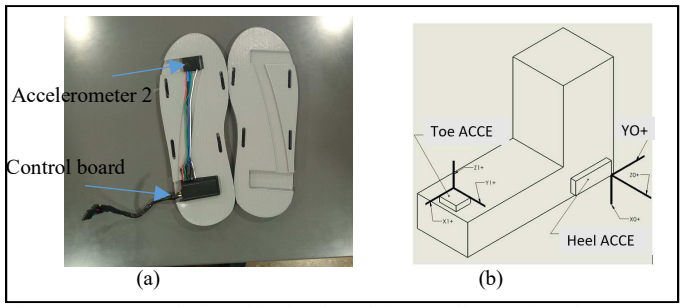

Fig. 1. (a) A 3D printing of the insole and (b) Coordiante orientation

B. Sensors and control System

The accelerometers and gyroscopes are installed to monitor motion and orientation changes. To measure the deformation of the foot, an array of four FBG sensors are attached at the bottom of insole as shown in Fig. 2. The FBG signals are collected via an FBG interrogator and visualized by a laptop. A miniaturized Arduino control board is used to collect signals received from accelerometers and gyroscope. Table 1 provides details about materials and systems used for experimental set up.

III. EXPRIMNETAL TESTING

A foot artificial skeleton is used to simulate different locomotion. Two monitors were setup to collect data from the FBG and the Arduino. The experimental set up is shown in Fig. 3.

Fig. 2. FBG sensors for monitroing foot defomation.

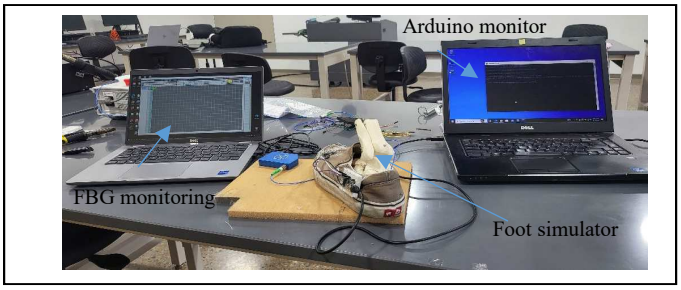

Fig. 3. Setup for data collection for the experimental testing.

TABLE 1. SENSOR TECHNOLOGY/DESCRIPTION USED IN THE PROTOTYPE SMART INSOLE

S/N	Table Column Head		
	Measured Parameter	***Sensor/Device***	***Model/Manufacturer***
1	Strain/Stress	FBG	Micronor Inc.
2	Acceleration	Accelerometer	Arduino Nano 33 BLE Sense Lite, Model: Nina-B306
3	Rotantional Angle	Gyroscope	Arduino Nano 33 BLE Sense Lite, Model: Nina-B306
4	Acceleation	Accelerometer	HiLetgo 3pcs GY-521, Model: MPU-6050

RESULTS AND DISCUSSIONS

Deformation Monitoring of the Foot

The deformation of the foot measured by strain measurement of FBG sensors is presented in Fig. 4. As it is realized, the strain follows a periodic pattern which represents the deformation of the foot during a normal walking cycle.

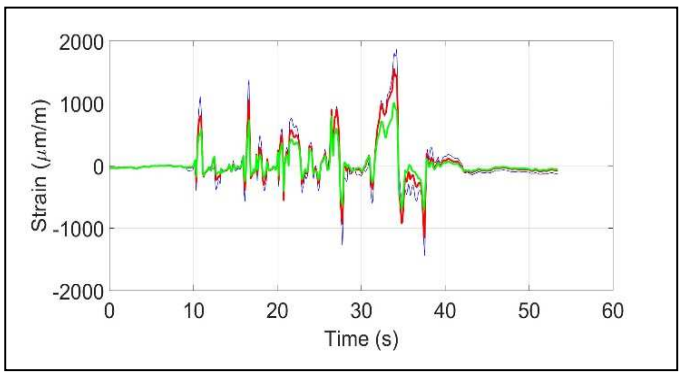

Fig. 4. Strain induced in insole during a regular walking cycle measured by FBG Sensors.

Monitoring of Dynamic Motion of the Foot

Fig. 5 represents a normal walking gait. At the first 4-7s, the accelerations remain relatively stable with minor fluctuations which shows that the person was in balanced mode without movement. From 7s onwards, the accelerations become more erratic, with visible spikes and troughs, indicating the dynamic nature of walking, including the heel strike and toe-off phases. After approximately 9 seconds, there is a significant spike in the blue (HeelAccel X) and other axes, indicating an irregular step.

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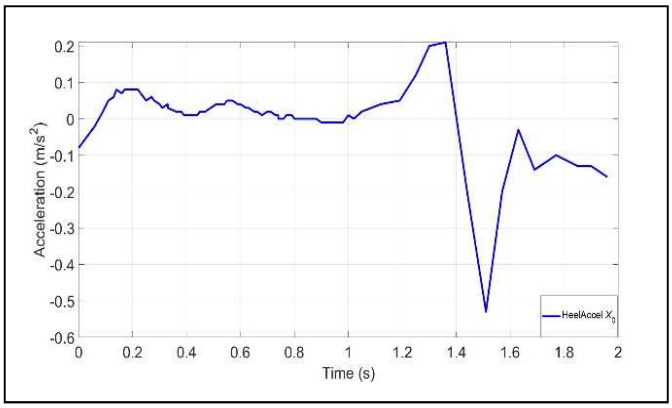

Fig. 5. Acceleration measured by two accelerometers attached at the side and front of the shoe during a normal walking.

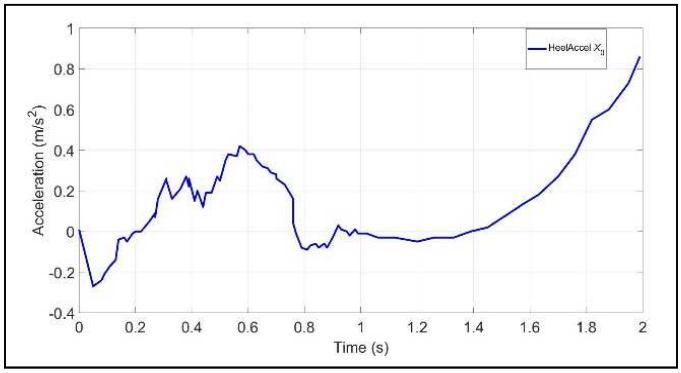

Fig. 6. Heel Acceleration in X_0 during one eversion cycle

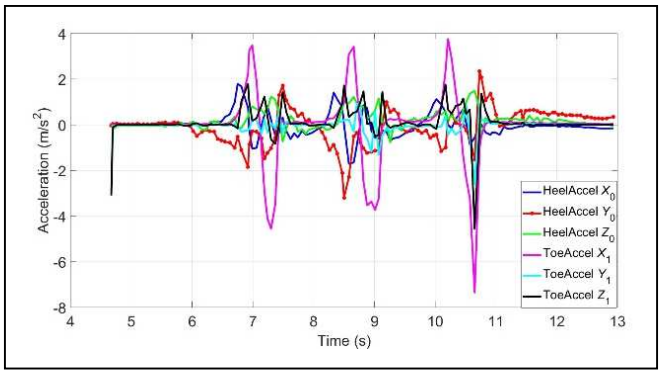

Fig. 7. Heel Acceleration in X_0 during one Inversion cycle

Figs. 6 and 7 shows the heel accelerations for eversion and inversion cycles for a foot drop issue. As it is realized the monitoring system accurately measure the gait cycles for different scenarios in real time.

CONCLUSIONS

A wearable insole has been integrated with multiple sensors has been developed to monitor locomotion of an individual. A wide range of human foot-ankle related activities have been investigated. As a specific application of the device, the walking cycle of an individual with foot-drop issue has been simulated.

It is realized that the wearable device can measure multiple locomotion parameters, including rotation angle, ground forces, and acceleration applied on the foot in real time with a high level of accuracy. The developed system will further be improved to design a shoe with embedded sensors, signal processing, and data transmission system

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